CO3 AT1-Case Study

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Case Study 1 – Detecting an Unusual Customer

Aim

To calculate the Z-score of a customer's spending and determine whether the customer's spending behavior is normal or unusually high compared with the average spending of all customers.

Introduction

In Data Science, organizations collect and analyze customer transaction data to understand purchasing behavior. Some customers spend much more or much less than the average customer. Such customers are called outliers or anomalies.

One of the most common statistical techniques used to identify unusual observations is the Z-score.

A Z-score indicates how many standard deviations a data value is away from the mean. It helps businesses detect abnormal customer behavior, fraudulent transactions, or high-value customers.

For example:

- Banking systems use Z-scores to detect suspicious transactions.
- E-commerce websites identify premium customers.
- Insurance companies detect fraudulent claims.
- Healthcare organizations identify abnormal medical readings.

Problem Statement

A shopping website observes that the average amount spent by customers is ₹2000, with a standard deviation of ₹400. A particular customer spends ₹3200.

The objectives are to:

1. Calculate the customer's Z-score.
2. Determine whether the spending is unusually high.
3. Explain the meaning of a negative Z-score.
4. Interpret another customer's Z-score of −1.5.

Theory

Mean (μ)

The mean is the average value of all observations.

It is calculated as

$$\mu = \frac{\sum X}{N}$$

where

- μ = Mean
- X = Observation
- N = Number of observations

Standard Deviation (σ)

Standard deviation measures how much the observations vary from the average value.

- Small standard deviation → Data is close to the mean.
- Large standard deviation → Data is spread out.

Z-score

The Z-score tells how far a value is from the mean in terms of standard deviations.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Given Data

Mean (μ) = ₹2000

Standard Deviation (σ) = ₹400

Customer Spending (X) = ₹3200

Step-by-Step Calculation

1. **Calculate the customer's Z-score.**

Write the formula

$$Z = \frac{X - \mu}{\sigma}$$

Substitute the values

$$Z = \frac{3200 - 2000}{400}$$

Subtract

$$3200 - 2000 = 1200$$

Therefore,

$$Z = \frac{1200}{400}$$

Divide

$$Z = 3$$

Answer

Customer's Z-score = 3

2. Determine whether the spending is unusually high.

Interpretation

A Z-score of 3 means the customer spends three standard deviations above the average spending.

Generally,

Z-score	Interpretation
0	Exactly average
+1	Slightly above average
+2	Much higher than average
+3	Extremely high spending
-1	Below average
-2	Much lower than average

Since the calculated Z-score is **3**, the customer's spending is **unusually high**.

The shopping website may classify this customer as:

- Premium customer
- High-value customer
- Potential VIP customer

3. Explain the meaning of a negative Z-score.

Meaning of a Negative Z-score

A negative Z-score indicates that the value lies below the average.

Example:

If a customer has a Z-score of -1, it means the customer spends one standard deviation less than the average.

Negative Z-scores are common and do not necessarily indicate a problem.

4. Interpret another customer's Z-score of −1.5.

Interpretation of Z-score = −1.5

Suppose another customer has

$$Z = -1.5$$

This means

- The customer spends 1.5 standard deviations below the average.
- The customer spends less than most customers.
- However, the spending is not considered an extreme outlier.

Applications of Z-score

The Z-score technique is widely used in Data Science.

Banking : Detect fraudulent transactions.

Healthcare : Identify abnormal blood pressure or sugar levels.

Stock Market : Detect unusual price changes.

Advantages

- Easy to calculate.
- Detects unusual observations.
- Useful in anomaly detection.

Result

The calculated Z-score is 3.

Therefore, the customer's spending is significantly higher than the average, indicating an unusual spending pattern. This customer can be classified as a high-value customer.

Conclusion

The Z-score is an effective statistical measure for detecting unusual observations in a dataset. In this case, the customer's spending is three standard deviations above the mean, indicating exceptionally high spending behavior. Such analysis helps organizations identify premium customers, detect anomalies, and make better business decisions. Z-score analysis is widely used in data science, finance, healthcare, and e-commerce because it provides a simple and reliable method for comparing individual observations with the overall dataset.

Case Study 2 – Delivery Time Analysis

Aim

To analyze the delivery times of an online delivery service using statistical measures such as Mean, Median, Quartiles, Interquartile Range (IQR), and Outlier Detection to understand delivery performance.

Introduction

Delivery time plays a significant role in customer satisfaction for online shopping platforms. Customers expect orders to be delivered quickly and consistently. Data Science helps companies analyze delivery performance using statistical methods.

Sometimes a few deliveries take much longer than others because of traffic, weather conditions, or other unexpected events. These unusually high or low values are called outliers.

To detect such values, Data Scientists use the Interquartile Range (IQR) method. It helps organizations identify abnormal observations and improve delivery efficiency.

Problem Statement

An online delivery service records the following delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

The objectives are to:

1. Calculate the Mean and Median.
2. Find Quartiles (Q1 and Q3).
3. Calculate the Interquartile Range (IQR).
4. Detect outliers using the 1.5 × IQR rule.
5. Determine whether the Median or Mean better represents the data.

Theory

Mean

The mean is the average of all observations.

$$\text{Mean} = \frac{\sum X}{N}$$

Median

The median is the middle value after arranging data in ascending order.

If the number of observations is even,

$$\text{Median} = \frac{\text{Middle Value 1} + \text{Middle Value 2}}{2}$$

Quartiles

Quartiles divide data into four equal parts.

- **Q1** = First Quartile (25%)
- **Q2** = Median (50%)
- **Q3** = Third Quartile (75%)

Interquartile Range (IQR)

IQR measures the spread of the middle 50% of the data.

$$IQR = Q3 - Q1$$

Outlier Detection

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

Any value outside these limits is considered an outlier.

Given Data

Delivery Times (minutes)

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Step-by-Step Calculation

1.Calculate the Mean and Median.

Calculate Mean

$$25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

Number of observations = 10

$$Mean = \frac{326}{10} = 32.6$$

Calculate Median

Since there are 10 observations,

Middle values are

30 and 30

$$Median = \frac{30 + 30}{2} = 30$$

2.Find Quartiles (Q1 and Q3).

Calculate Quartiles

Lower Half

25, 27, 28, 29, 30

Q1 = 28

Upper Half

30, 31, 32, 34, 60

Q3 = 32

3.Calculate the Interquartile Range (IQR).

Calculate IQR

$$IQR = 32 - 28 = 4$$

Lower Boundary

$$28 - 1.5(4)$$
$$28 - 6 = 22$$

Upper Boundary

$$32 + 1.5(4)$$
$$32 + 6 = 38$$

4.Detect outliers using the 1.5 × IQR rule.

Detect Outliers

Since

60 > 38

Therefore,

60 s an outlier.

5.Determine whether the Median or Mean better represents the data.

Interpretation

Mean = 32.6 minutes

Median = 30 minutes

The mean is affected by the unusually large value (60 minutes).

The median remains unchanged.

Therefore,

Median better represents the typical delivery time.

Applications

- Delivery Performance Analysis
- Logistics Management
- Food Delivery Services
- Courier Companies
- Warehouse Operations

Advantages

- Detects abnormal delivery times.
- Improves logistics planning.
- Helps increase customer satisfaction.
- Easy statistical method.
- Useful in real-time monitoring.

Result

Mean = **32.6**

Median = **30**

Q1 = **28**

Q3 = **32**

IQR = **4**

Outlier = **60 minutes**

The median is a better measure because the dataset contains an outlier.

Conclusion

The Interquartile Range (IQR) method is an effective technique for identifying outliers in a dataset. In this case, the delivery time of 60 minutes is an outlier because it exceeds the upper limit of 38 minutes. Since the outlier affects the mean, the median provides a more accurate representation of the typical delivery time. This analysis helps delivery companies monitor performance, detect delays, and improve customer satisfaction.

Case Study 3 – Comparing Temperature Variations of Two Cities

Aim

To compare the temperature variations of two cities by calculating the Mean, Variance, and Standard Deviation and determine which city has more consistent temperatures.

Introduction

Weather data is collected every day by meteorological departments to study climate patterns. Data scientists analyze temperature records to understand climate changes, seasonal variations, and weather consistency.

Statistical measures such as Mean, Variance, and Standard Deviation help determine how much temperatures vary over time. A city with a lower standard deviation experiences more stable weather, while a city with a higher standard deviation has greater fluctuations.

This analysis is useful in agriculture, tourism, environmental science, and disaster management.

Problem Statement

The average daily temperatures (°C) of two cities for one week are given.

City A: 28, 29, 30, 30, 31, 30, 32

City B: 22, 26, 29, 31, 35, 37, 40

The objectives are to:

1. Calculate the mean temperature of both cities.
2. Calculate the variance.
3. Calculate the standard deviation.
4. Compare temperature consistency.
5. Identify which city has more temperature variation.

Theory

Mean

The mean is the average value.

$$\bar{x} = \frac{\sum X}{N}$$

Variance

Variance measures how far the observations are spread from the mean.

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{N}$$

Standard Deviation

Standard deviation is the square root of variance.

$$\sigma = \sqrt{\sigma^2}$$

Given Data

City A

28, 29, 30, 30, 31, 30, 32

City B

22, 26, 29, 31, 35, 37, 40

Step-by-Step Calculation

1.Calculate the mean temperature of both cities.

Mean of City A

$$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$$

Number of observations = 7

$$Mean = \frac{210}{7} = 30°C$$

Mean of City B

1. $22 + 26 + 29 + 31 + 35 + 37 + 40 = 220$

$$Mean = \frac{220}{7} = 31.43°C$$

***2.*Calculate the variance.**

Variance

Calculate $(x - \bar{x})^2$ for each value and divide the total by the number of observations.

City A has a small variance because temperatures are close to the mean.

City B has a larger variance because temperatures vary widely.

3.Calculate the standard deviation.

Standard Deviation

Take the square root of the variance.

A smaller standard deviation indicates more consistent temperatures.

A larger standard deviation indicates greater variation.

4.Compare temperature consistency.

Interpretation

- City A has temperatures close to its average.
- City B has larger fluctuations.

5.Identify which city has more temperature variation.

- Therefore, City A experiences more stable weather conditions.

Applications

- Weather Forecasting
- Agriculture Planning
- Climate Research
- Disaster Management
- Environmental Monitoring

Advantages

- Measures data consistency.
- Easy comparison between datasets.
- Helps identify climate variations.
- Supports weather prediction.
- Useful in scientific research.

Result

The mean temperatures are calculated, and City A shows lower variance and standard deviation. Therefore, City A has more consistent temperatures, while City B experiences greater temperature variation.

Conclusion

Mean, variance, and standard deviation are essential statistical measures for analyzing temperature data. They help determine the average temperature and the degree of variation. In this case, City A has more stable weather conditions, whereas City B shows greater fluctuations.

Case Study 4 – Data Center CPU Usage Analysis

Aim

To analyze the CPU utilization data using descriptive statistics such as Mean, Median, Mode, Range, Quartiles, Interquartile Range (IQR), and Outlier Detection to evaluate system performance and identify abnormal CPU usage.

Introduction

CPU utilization is an important performance metric in a data center. Monitoring CPU usage helps administrators ensure that servers are operating efficiently. By using descriptive statistics, we can summarize the data and identify unusual CPU usage patterns. The Interquartile Range (IQR) method is commonly used to detect outliers, which may indicate heavy workloads or system issues.

Problem Statement

A data center records the following CPU utilization percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Calculate the Mean, Median, Mode, Range, Quartiles, IQR, identify any outliers, and comment on the skewness of the dataset.

Formula

- Mean = ΣX / N
- Median = Middle value of the ordered dataset
- Mode = Most frequently occurring value
- Range = Maximum − Minimum
- IQR = Q3 − Q1
- Upper Limit = Q3 + 1.5 × IQR

Calculation

1.Calculate the Mean, Median, Mode

Mean

$$\frac{40+42+45+45+46+48+50+52+55+90}{10} = \frac{513}{10} = 51.3$$

Median

$$\frac{46+48}{2} = 47$$

Mode 45

2.Calculate the Range

Range

$$90 - 40 = 50$$

3.Find Q1,Q3 and IQR

Quartiles

- Q1 = 45
- Q3 = 52

Interquartile Range (IQR)

$$IQR = 52 - 45 = 7$$

4.Identify possible outliers.

Outlier Detection

Upper Limit

$$52 + 1.5(7) = 62.5$$

Since 90 > 62.5, the value 90 is an outlier.

5.Determine the direction of skewness.

Skewness

The presence of the high value 90 increases the mean, making the data positively skewed (right-skewed).

Interpretation

Most CPU utilization values lie between 40% and 55%, indicating normal system performance. However, the value 90% is much higher than the other observations and is identified as an outlier. This unusually high CPU utilization may be caused by heavy workloads, excessive user requests, hardware issues, or software failures.

Applications

- Server Performance Monitoring
- Cloud Computing
- Network Management
- Resource Allocation

- Performance Optimization

Advantages

- Summarizes CPU utilization effectively.
- Detects abnormal CPU usage.
- Helps optimize system performance.
- Supports better resource management.
- Assists in identifying performance bottlenecks.

Result

The statistical analysis of CPU utilization produced the following results:

- Mean = 51.3%
- Median = 47%
- Mode = 45%
- Range = 50%
- Q1 = 45%
- Q3 = 52%
- IQR = 7%
- Outlier = 90%
- Distribution = Positively Skewed

The value 90% is identified as an outlier because it exceeds the upper limit calculated using the 1.5 × IQR rule, indicating unusually high CPU usage.

Conclusion

Descriptive statistical analysis helps evaluate CPU utilization and identify abnormal system behavior. In this dataset, the CPU usage value of 90% is detected as an outlier, indicating unusually high system load. Such analysis enables administrators to monitor system performance, detect anomalies, optimize resources, and improve the efficiency and reliability of data center operations.

Case Study 5 – Comparing Two Datasets

Aim

To compare two datasets using Mean, Variance, and Standard Deviation and determine which dataset is more consistent.

Introduction

In Data Science, comparing datasets helps us understand not only their average values but also how the data is distributed. Two datasets may have the same mean but different levels of variation. Therefore, statistical measures such as Variance and Standard Deviation are used to measure the consistency and spread of the data. These measures help in making accurate decisions in business, healthcare, finance, and quality control.

Problem Statement

Consider the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

Calculate the Mean, Variance, and Standard Deviation for both datasets and determine which dataset is more consistent. Also explain why the mean alone is not sufficient for comparison.

Formula

Mean

$$\bar{x} = \frac{\sum X}{N}$$

Variance

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{N}$$

Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

Calculation

1.Calculate the mean of both datasets.

Mean of Dataset A

$$\frac{10 + 20 + 30 + 40 + 50}{5} = \frac{150}{5} = 30$$

Mean of Dataset B

$$\frac{28 + 29 + 30 + 31 + 32}{5} = \frac{150}{5} = 30$$

Both datasets have the same mean (30).

2.Calculate the variance of both.

Variance

Dataset A

$$Variance = \frac{400 + 100 + 0 + 100 + 400}{5} = \frac{1000}{5} = 200$$

Dataset B

$$Variance = \frac{4 + 1 + 0 + 1 + 4}{5} = \frac{10}{5} = 2$$

3.Calculate the standard deviation of both.

Standard Deviation

Dataset A

$$SD = \sqrt{200} = 14.14$$

Dataset B

$$SD = \sqrt{2} = 1.41$$

4. Which dataset has greater spread?
Dataset A has the greater spread because its values are widely distributed from 10 to 50. It has a higher variance (200) and higher standard deviation (14.14) than Dataset B. Therefore, Dataset A shows greater variability.

5. If both datasets represent model errors, which model appears more consistent?
Dataset B appears to be more consistent because it has a lower variance (2) and lower standard deviation (1.41). The error values are closely clustered around the mean, indicating that the model produces more stable and reliable predictions.

6. Explain why mean alone is not enough to compare datasets.
The mean only shows the average value of a dataset and does not indicate how the data is spread. Two datasets can have the same mean but very different distributions. Variance and standard deviation measure the spread and consistency of the data. Therefore, these measures are essential for comparing datasets accurately. In this case, both datasets have the same mean (30), but Dataset B is more consistent because its values are much closer to the mean.

Applications

- Business Data Analysis

- Financial Forecasting
- Quality Control
- Healthcare Analytics
- Educational Performance Analysis
- Scientific Research

Advantages

- Helps compare the consistency of datasets.
- Measures the spread of data accurately.
- Supports better decision-making.
- Identifies stable and reliable datasets.
- Useful for predictive analysis and statistical reporting.

Result

The statistical comparison of the two datasets produced the following results:

Measure	**Dataset A**	**Dataset B**
Mean	30	30
Variance	200	2
Standard Deviation	14.14	1.41

Although both datasets have the same mean (30), Dataset B has a much lower variance and standard deviation, indicating that its values are more closely clustered around the mean. Therefore, Dataset B is more consistent, while Dataset A shows greater variability.

Conclusion

Mean alone is not sufficient to compare datasets because it only represents the average value. Variance and standard deviation provide information about the spread and consistency of the data. In this case, Dataset B is more stable and consistent because it has a much lower variance and standard deviation than Dataset A. These statistical measures are widely used in Data Science to compare datasets and support informed decision-making